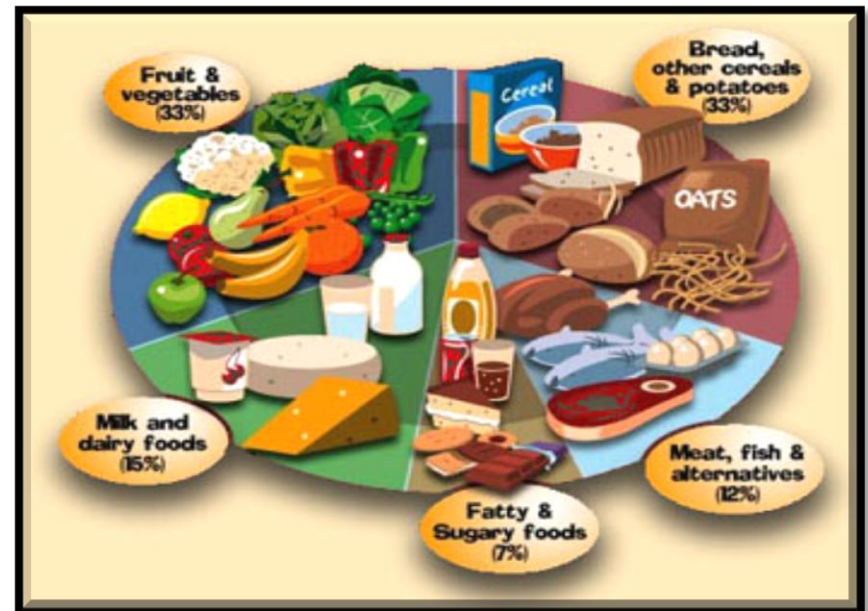
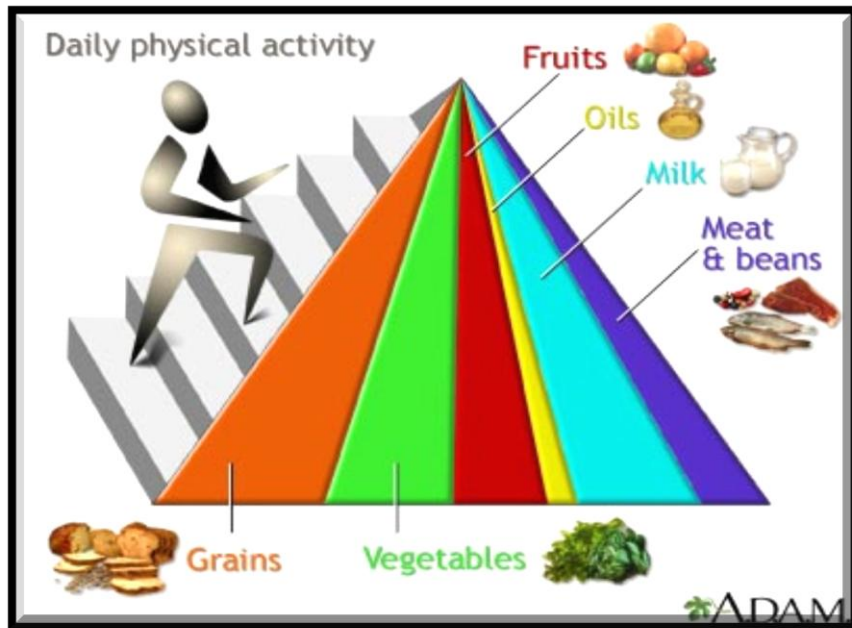


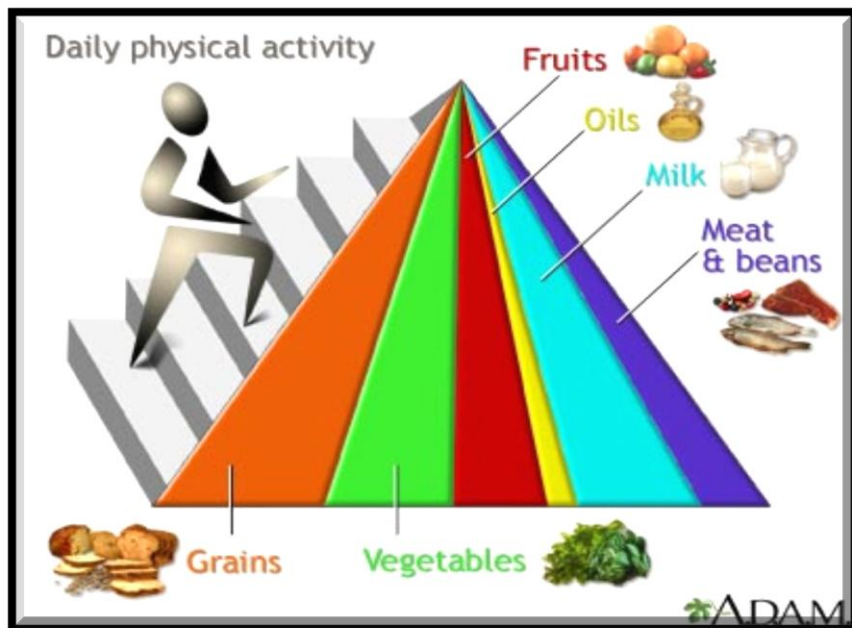
NUTRITIONAL NEEDS AND MEAL PLANNING

4th Level, 1437H



NUTRITIONAL NEEDS AND MEAL PLANNING

L1 : L4



LECTURE 1

FEEDBACK

Definition of meal planning

General Objectives of the course

After study this cores we should be able to

- 1- Describe the concept and aim of meal planning .
- 2- Understand the different factors which influence meal planning .
- 3- Identify foods which must be included while planning nutritionally balanced meals for different age groups .

General Objectives of the course

- 4- Relate the RDA nutritive value of meals served in catering establishments .
- 5- Evaluate the nutritional adequacy of meals served to individuals .
- 6- Understand the special requirements for nutrients at various stages in life and how these can be met .

General Objectives of the course

- 7- Define the term balanced diet .
- 8- Understand the importance of consuming a balanced diet .
- 9- Explain the basis for dividing food into groups
- 10- Discuss the use of food groups and RDA in planning balanced diet .

WHAT IS MEAL PLANNING

Meal planning is making a plan of meals with adequate nutrition for every member of the family within the available resources. The term 'available resources' means whatever the family has in terms of time, energy and money.

IMPORTANCE OF MEAL PLANNING

- 1-meeting the nutritional requirements of the family members.
- 2-It helps us to decide what to eat each day and in each meal. We can call it our 'daily food guide'.

The aim of menu planning

- (1) Provide the nutritional requirements of the family members
- (2) make the food economical
- (3) cater to the food preferences of individual members
- (4) save energy, time and money
- (5) use left over food
- (6) Provide attractive , appetizing meals with no monotony

Definitions

Meal:	The food served and eaten <u>especially</u> at one of the customary , regular occasions for taking food during the day , as breakfast , lunch or supper.
Menu:	A list of all the kinds of food that are available for a meal.
Diet :	A particular selection of food, especially designed or prescribed to improve a person's physical condition or to prevent or treat a disease.
Dish:	Food prepared in a particular way as part of a meal.

Factors affecting meal planning

1. 1. Nutritional requirements (Nutritional Adequacy)
2. Age
3. Sex
4. Physical Activity
5. Economic
- Time, energy and skill considerations Considerations
7. Seasonal availability
8. Religion, region, cultural patterns, traditions and customs
- 9- Variety in color and texture
10. Likes and dislikes of individuals
11. Satiety Value

Factors affecting meal planning

- 1. Nutritional requirements (Nutritional Adequacy)** This is the most important factor
- **Nutritional requirements** means that the nutritional requirements of all the family members are fulfilled.
- For example, you know a growing child needs more protein, a pregnant or lactating woman needs calcium, etc.

Factors affecting meal planning

2. Age

People normally eat according to their age.

You must have observed in your family that the diet of various members of different age groups differs in quantity.

A new born baby drinks only milk, a small child's meal is also of very small quantity,

an adolescent eats still more in amount and variety of foods.

Similarly,

you must have seen your grandfather eating less food and also that they prefer soft and easy to digest foods.

Factors affecting meal planning

3. Sex

Another factor which determines the dietary intake.

Dietary requirement of female adolescent and adult males are **more than** their counterparts.

4. Physical Activity

The kind of work a person does affects the kind and amount of food they need to take.

4. Physical Activity

Activity Multiplier:

Sedentary = $\text{BMR} \times 1.2$ (little or no exercise, desk job)

Lightly active = $\text{BMR} \times 1.3$ (light exercise/ sports 1-3 days/week)

Moderately active = $\text{BMR} \times 1.5$ (moderate exercise/ sports 6-7 days/week)

Very active = $\text{BMR} \times 1.7$ (hard exercise every day, or exercising 2 xs/day)

Extra active = $\text{BMR} \times 1.9$ (hard exercise 2 or more times per day, or training for marathon, or triathlon, etc.)

Factors affecting meal planning

5. Economic Considerations

Money available to the family to be spent on food is another major factor.

Foods like milk, cheese, meat, fruits, nuts etc. are **expensive**.

However, alternative sources like toned milk, seasonal fruits and vegetables are **less costly** and at the same time nutritious.

6. Time, energy and skill considerations

While planning the meals, you should consider the resources like time, energy and skill available to the family.

- Having little time does not mean your meals need to be “fast food”.
- Plan ahead or use convenience foods.
 - Precut fruit/vegetables
 - Salad mixes
- Microwaves, dishwashers and food processors can be time saving tools!



Factors affecting meal planning

7. Seasonal availability

Some foods are available in summers while some in winters. The off season foods are expensive and less nutritious, while those in season are fresh, nutritious, tasty and cheap. Hence, while planning seasonal foods should be used.

8. Religion, region, cultural patterns, traditions and customs

Religious beliefs prevalent in the family also have an influence.

For example,

**if you are a vegetarian, your diet will not have any meat or meat product,

**Hindus do not eat beef

**Muslims do not eat pork.

Factors affecting meal planning

9. Variety in color and texture

.These factors help you to make meals more appealing, attractive and hence more acceptable

10. Likes and dislikes of individuals

The food you serve should cater to the likes and dislikes of the individual family members. It is often better to change the form of some particularly nutritious food item, rather than omitting it completely. For example, if someone in your family does not like milk, you can give it in the form .of curd

Factors affecting meal planning

11. Satiety Value

While planning meals, take care that you select foods which provide satiety value. Meals which produce inadequate satiety, will lead to onset of hunger pangs, which in turn will affect the working capacity and efficiency of a person.

Categories of menu pricing styles

1-A' la Carté:

2-Table d' Hôte:

3-Cyclic Menu:

Categories of menu pricing styles

1-A' la Carté:

This is a French word and means “**According to the Card or Customer’s Order**”
It’s a list of all the dishes available in a particular restaurant. In this menu, the guests may select separately to compile his or her menu and is required to pay only for the food or beverage items ordered

2-Table d' Hôte:

This is a French word and means “**Table of the Host**”

In this, the meal is provided at a fixed price and the customer has to pay for the complete meal no matter him/her consuming the whole meal or not.

A limited choice may be given to the customer for each course or between courses.

Categories of menu pricing styles

3-Cyclic Menu:

It's a menu which changes every day for a certain set of days before the cycle ends. In this the menu changes every day for a particular week. After the completion of the cycle, a new set of menu is prepared and followed, keeping in mind the seasonal availability of dishes.

Advantages: -

- * Effective cost control
- ** Stocking of unnecessary goods is avoided.
- *** Less storage space for perishables is required.
- **** Less food wastage.
- ***** Well planned labor requirement
- ***** Fewer cooking equipment required.

Disadvantages: -

- * Choice for the customer is not there.
- ** Frequency of repetition of dishes is higher
- *** Extensive knowledge is required for the preparation.

LECTURE 2

Balanced & healthy Diet

WHAT IS BALANCED DIET?

A balanced diet means getting the right types and amounts of foods and drinks to supply nutrition and energy for maintaining body cells, tissues, and organs, and for supporting normal growth and development.

A balanced diet needs to contain foods from all the main food groups in the correct proportions to provide the body with optimum nutrition.

WHAT ARE THE COMPONENTS OF A BALANCED DIET?

A balanced diet must provide the building blocks for growth and repair, and sufficient energy. There are seven components to a balanced diet:

1- Carbohydrates

2- Proteins

3-Fats

4- Fibers

5-Water

6- Vitamins

7-Minerals

Characteristics of a Balanced Diet

A balanced diet contains both plant and animal foods and fulfills following requirements:

- *meets the nutritional requirements of an individual**
- *includes foods from all the food groups**
- *contains a variety of foods**
- *consists of seasonal foods**
- * It suits the taste and meets the desires of the individual eating .**

Steps In Planning Balanced Meal

1- Collect information regarding the customer with respect to .

A- Age B- Gender C- Activity level .

D- Religion E- Socio –economic background . F-Food habits

2 –Check the RDAs for energy and proteins .

3- Prepare a food plan , list number of servings from each food group to meet the RDA .

4- Decide on number of meals.

5- Distribute servings for each meals .

6- Select foods within each group and state their amount.

7- Plan a menu .

8- Cross check to ensure that all food groups are included in requisite amounts.

Why Is It Important to Have a Balanced Diet?

1. It reduced risk for illness and Infections, reduce the risk of chronic diseases like cancer , high blood pressure ,heart disease and diabetes.
2. It Helps Control Weight
3. It Promotes Healthy Body Growth
4. It Helps Promote Mental Health

Healthy Eating



A healthy diet

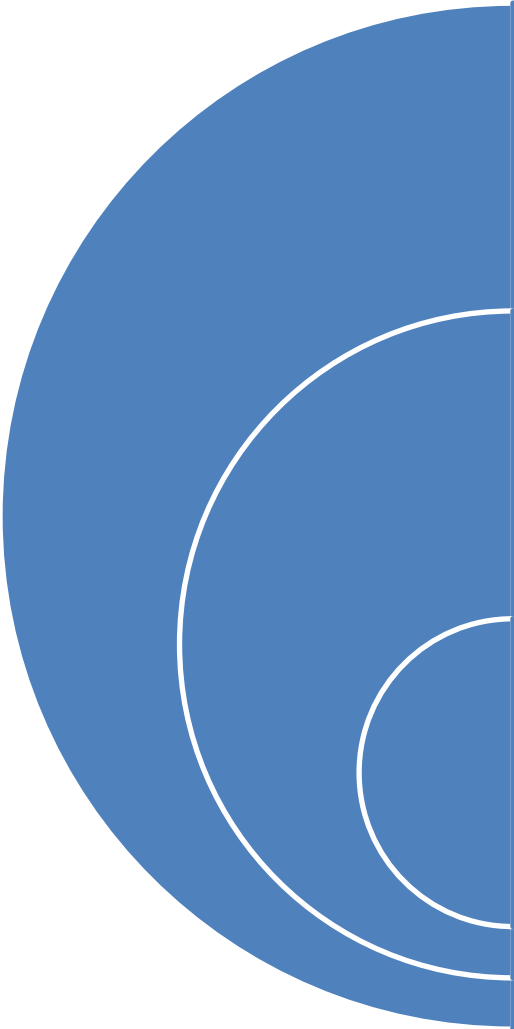
- is one that provides the right amount of energy to keep weight in the desirable range; the proper types and balance of carbohydrates, proteins, and fats; plenty of water; and sufficient but not excessive amounts of essential vitamins and minerals.

- -A healthy diet is rich in nutrient-dense foods.

Nutrient density

- is a measure of the nutrients a food provides compared to its energy content.
- (Nutrient-dense foods contain more nutrients per calories).

The principles of making healthy diet



1-Variety

2-Balance

3- Moderation

The principles of making healthy diet

1-Variety

Eat a Variety of Foods, no one food can provide all the nutrients the body needs for optimal health. Eating a variety of foods, however, helps ensure an adequate nutrient intake. Variety means including grains, vegetables, fruits, dairy products, and meats in the diet. Some of these foods are rich in protein and minerals, others in vitamins and phytochemicals .

Variety means choosing many different foods from within each food group.

For instance, if you always choose red meat as a protein source you will be missing out on the fiber in beans .

Variety comes not only from choosing different foods each day, but also each week, and each season. If you had apples and grapes today, have blueberries and cantaloupe tomorrow.

The principles of making healthy diet

Choosing a varied diet is also important because there are interactions between different foods and nutrients . These interactions may be positive, enhancing nutrient utilization, or negative, inhibiting nutrient use.

For example, consuming iron with orange juice enhances iron absorption, while consuming iron with milk may reduce its absorption.

The principles of making healthy diet

2-Balance

Balance involves mixing and matching foods in proportions that allow you to get enough of the nutrients you need and not too much of ones that might harm your health.

For example ,when you choose a food that is high in fat, like chocolate, then balance that choice with a low-fat one, like a piece of fruit.

-It also balances the energy taken in with the energy used up in daily activities so body weight stays in the healthy range

The principles of making healthy diet

3- Moderation

Moderation means watching your portions and passing up the super sizes.

-Its also means not consuming too much energy, too much fat, too much sugar, too much salt, or too much alcohol.

Choosing moderately will help you maintain a healthy weight and help prevent some of the chronic diseases like heart disease and cancer.

Eight tips for eating well

- 1-Base your meals on starchy foods: ***should make up $\frac{1}{3}$ of our diet**
- 2-Eat lots of fruits and vegetables : ***eat at least 5 portions of a variety of fruit and vegetables every day**
- 3-Eat more fish: **eat 2 portions of fish a week ,one of which should be oily.**
- 4-Cut down on saturated fat and sugar
- 5-Try to eat less salt (not more than 6g /day)
- 6-Get active
- 7-Drink plenty of water
- 8-Try not to skip breakfast:

Lecture 3

Consumer Food Habits And Preferences In Relation To Nutrition

Nutritional intake is largely dependent on *(food habits and preferences)* which related to several complex beliefs and habits which are very hard to change since they are acquired over a long period of time .

The intake of nutrients depends on what foods are preferred, selected , and consumed .

1- Food habits

- defined as the way in which individuals in response to social and cultural pressures, select , consume ,and utilize the available food supply .

2- Food preferences

- defined as the selection of food items from among a variety of acceptable foods .

Food preferences are based on

1- Sensory

2- Social

3- Psychological

4- Religious

5- Emotional

6- Cultural

7- Health

8- Economic

9- Preparation methods and other related factors .

THE FACTORS WHICH INFLUENCE ON FOOD INTAKE

- 1- Intrinsic Factors .**
- 2- Extrinsic Factors .**
- 3- Biological, Physiological, and Psychological Factors .**
- 4- Personal Factors .**
- 5- Socioeconomic Factors .**
- 6- Cultural And Religious Factors .**
- 7- Food Characteristics .**

THE FACTORS INFLUENCING FOOD INTAKE

1- Intrinsic Factors :

Intrinsic factors include certain influences directly associated with foods such as (appearances , color , odor , texture , temperature , flavor ,and quality).

- The way food is presented (including both desirable and undesirable attributes), the way it is arranged on the plate , and the temperature at which it is served all have an impact on food preferences

THE FACTORS INFLUENCING FOOD INTAKE

2- Extrinsic Factors :

- Including direct external factors such as :

A- Environment

B- Advertising

C- Time and seasonal variations :

THE FACTORS INFLUENCING FOOD INTAKE

2- Extrinsic Factors

A- Environment :

-Food preferences are affected by the environment in which food is served such as :

A- Homes

B- Restaurants

C- Nursing homes

D- Clubs

E- Hospital which have different effect on the selection of food .

THE FACTORS INFLUENCING FOOD INTAKE

2- Extrinsic Factors

B- Advertising

Many food service operations use advertisements and promotions to attract customers .

C- Time and seasonal variations :

Food selections appear to be some what immune to the influence of certain natural phenomena , such as seasonal variations , outdoor ,temperatures and the day of the week.

For example (watermelons are preferred during the hot summer months ,and hot chocolate is a winter favorite) .

TEMPERATURE



THE FACTORS INFLUENCING FOOD INTAKE

3- Biological, Physiological, and Psychological Factors :

- These factors are each broad in scope but are grouped together here because they are closely interrelated.
 - Physiological disorders can have a profound effect on food preferences by changing food appreciation, perception, or appetite.
-
- *People with food allergies, digestive problems, and dental disease are limited in the foods that are safe and comfortable for them to eat.
 - *People are depressed turn to chocolate or overeat in general while others eat less or stop eating.

THE FACTORS INFLUENCING FOOD INTAKE

4- Personal Factors

- 1- Level of expectation .**
- 2- Priority .**
- 3- Familiarity .**
- 4- Influence of other persons .**
- 5- Emotions and moods .**
- 6- Family unit .**
- 7- Educational status .**

THE FACTORS INFLUENCING FOOD INTAKE

5- Socioeconomic Factors

Socioeconomic factors such as income level, living conditions, and lifestyle as well as education affect the types and amounts of foods that are available.

-Individuals with limited incomes can choose only the types and amounts of foods that they can afford.

THE FACTORS INFLUENCING FOOD INTAKE

6- Cultural And Religious Factors

- Cultural and religious factors influences on food preferences may be transmitted from one generation to another .**
- Various religious restrictions have resulted in stable and rigid food preferences .**

For example .. Muslims and Jews prohibit the use of pork and pork products .

THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

Characteristics of foods including their sensory properties play an important role in their acceptance .

Food Characteristics such as

- 1- Color
- 2- Texture and shape
- 3- Consistency
- 4- Flavor
- 5- Method of food preparation
- 6- Serving temperature

THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

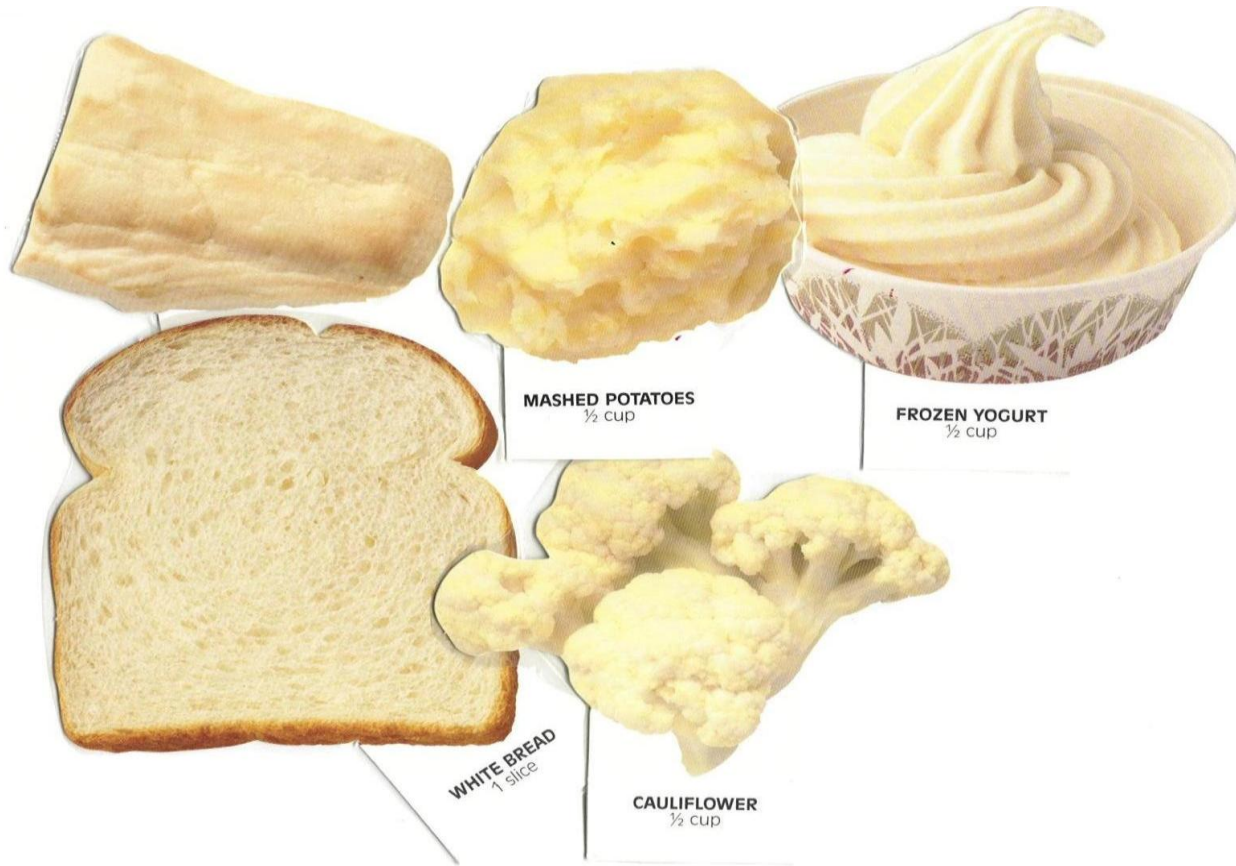
1- Color :

- Interesting and coordinating color combinations promote the acceptance of food and indirectly help stimulate appetite .
- Planning and arranging foods so that there is a good color combination on a plate , tray , counter or salad bar is important .

-Color has emphasizes the variety available for selection and a psychological impact on consumers . Red , orange , peach , pink , brown ,yellow , and light green are considered desirable food colors , purple, violet , dark green , gray , and olive less so .



COLOR



THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

2- Texture and shape :

-The texture and shape of foods also affect consumers preferences

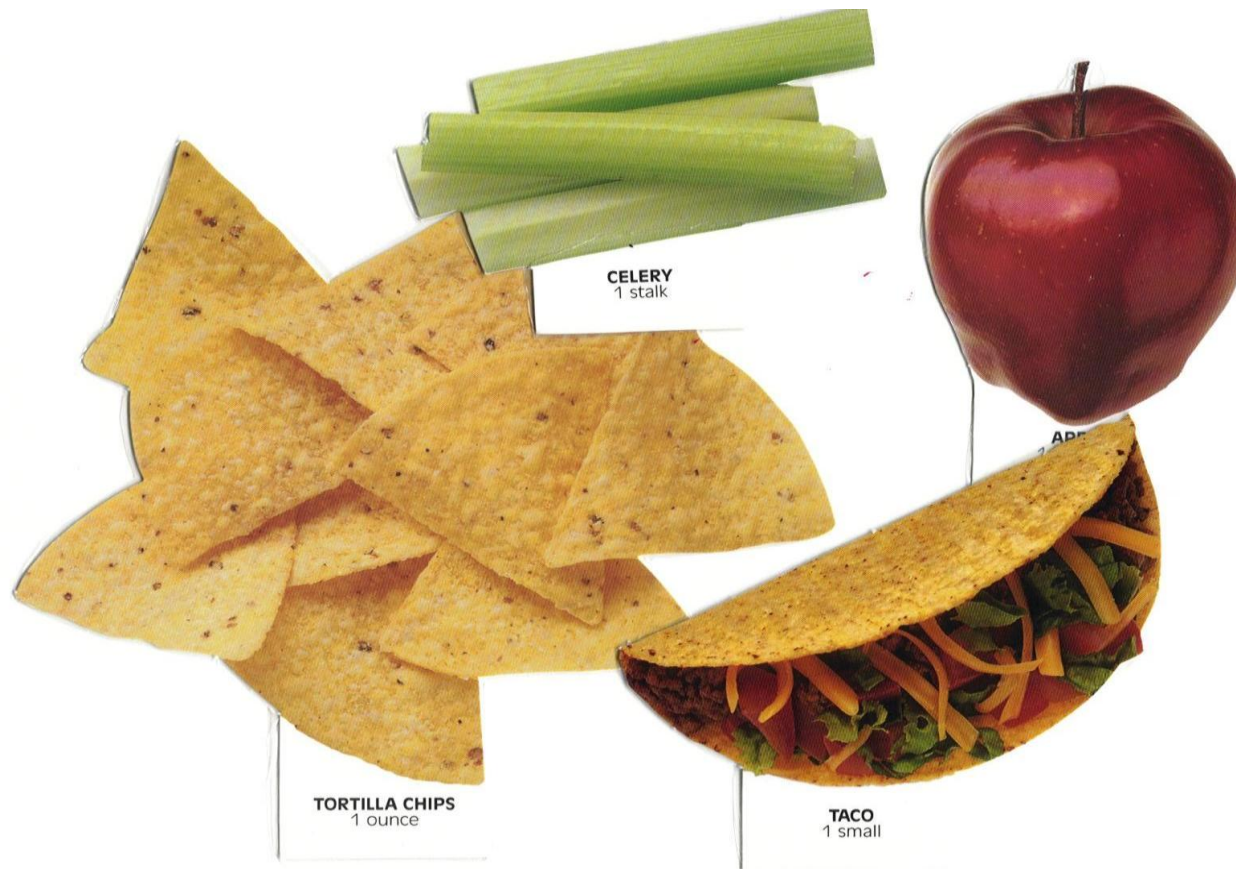
Certain foods are preferred because of their hard or soft texture .

-A desirable combination of soft and hard texture menu items is essential .

-Impressions of the texture and shape of a particular food are formed even before it is tasted .

-Texture can best be detected by mouth feel ., Soft ,hard , crunchy , chewy smooth , brittle , and grainy are some of the adjectives used to describe food texture .

TEXTURE



THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

3- Consistency :

-Consistency refers to the degree of viscosity or density of a product. Like texture , consistency provides variety among food choices , Runny , gelatinous , pasty , thin , thick , sticky , and gummy are the most common adjectives used to describe consistency .



THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

4- Flavor :

- Flavor has an important impact on food preferences. Food can have sweet , sour , bitter or salty flavors which can be present alone or in combination .
- In addition there are various off flavors and undesirable flavors that hinder the selection of foods, Broccoli cabbage and onions which provide strong flavors and have special nutrient contents , add to the flavor of meals .

FLAVOR



POTATO CHIPS
1 ounce

PICKLE
1 pickle



CHICKEN NOODLE SOUP

SNACK CRACKERS
4 crackers

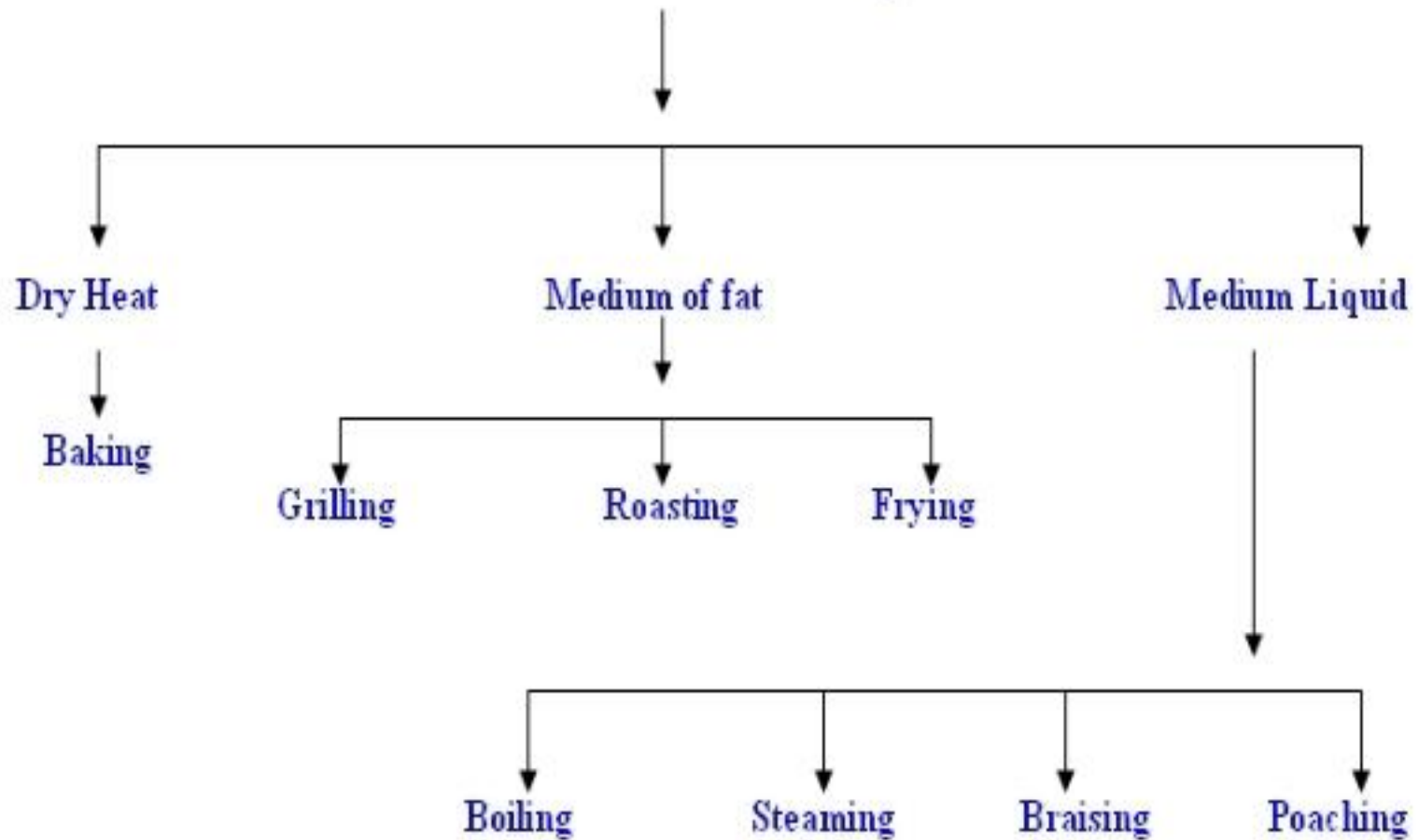
THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

5- Method of food preparation :

- Food can be prepared in many ways and variety in preparation is preferred . Methods commonly used include ... frying, baking , broiling , boiling , steaming , grilling , braising or a combination of these types .
- Each of these methods has an impact on food preferences as well as on the nutritional content of foods .

Method of Cooking



METHODS OF COOKING

- **Moist Methods:** Boiling, Simmering, Poaching, Stewing, Braising, Steaming, Pressure cooking.
- **Dry Methods:** Baking, Grilling, Roasting, Barbecuing.
- **Frying:** Dry, Shallow, Deep -fat, Stir.
- **Microwave cooking.**

CHOOSING A COOKING METHOD

- Ingredients available e.g. stewing beef or sirloin steak.
- Experience of cook.
- Time available.
- Equipment available.
- Desired result e.g. crispy, brown, moist.
- Fuel economy.
- Retention of nutrients.

BOILING / SIMMERING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking by convection & conduction in liquid @ 100°C in covered saucepan on the hob. •Simmering - 90 °C	•100 °C @ all times. •Vegetables - small amount of liquid. •Use cooking liquid - soups, sauces. •Cook food for shortest possible time.	•Meat - bacon, mutton. •Eggs. •Pasta. •Rice. •Veg.	•Needs little attention. •Clean. •Food stays moist. •Digestible. •Less greasy. •Little preparation. •Little flavour added. •Loss of nutrients. •Lacks texture.



POACHING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking by convection & conduction in liquid @ 85°C in covered container on hob or in oven. 	•85 °C @ all times. • Water barely moving, not bubbling. • Use for delicate foods that need gentle cooking.	•Fish. •De-shelled eggs. •Fruit.	•Clean. •Food stays moist. •Digestible. •Less greasy. •Little flavour added. •Loss of nutrients. •Lacks texture. •Slow. •Needs attention.

BRAISING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking meat in a small amount of stock, on a layer of vegetables (mirepoix). Meat is in a covered saucepan on the hob. •Combination of steaming and stewing.	•Use only enough stock to cover vegetables. •Food can be browned under grill before serving.	•Root vegetables •Chicken. •Offal. •Tougher cuts of meat.	•Little attention needed. •Food stays moist. •Complete meal in one pot - saves fuel & wash-up. •Less nutrient loss. •Digestible. •A lot of prep. •Slow. •Lacks texture.



STEWING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
• Slowly cooking food in a little liquid, by conduction & convection, using gentle heat (80 °C - 90 °C), in a covered container, on hob or in oven.	• Bring to boil & then reduce heat. • Temperature of 80 °C - 90 °C. • Keep covered. • Use pressure cooker to reduce time.	• Tough cuts of meat/fish. • Veg. • Fruit	• Little attention needed. • Food stays moist. • Complete meal in one pot - saves fuel & wash-up. • Less nutrient loss. • Digestible. • A lot of prep. • Slow. • Lacks texture



STEAMING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking food slowly in steam rising from boiling water. •Food can be steamed: •a. Between two plates. •b. In a covered bowl in a saucepan of boiling water. •c. In a steamer over boiling water. •d. On a trivet in a pressure cooker.	• A tightly fitting lid prevents evaporation. •Water boiling before & during cooking. •Food must not touch liquid.	•Thin pieces of chicken and fish. •Steamed puddings, e.g. canary, Christmas, chocolate. •Vegetables.	•No loss of nutrients. •Clean. •Food stays moist. •Digestible. •Not greasy. •Little flavour added. •Lacks texture. •Slow. •Not suitable for large pieces of food.



BAKING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
• Dry method of cooking food by convection currents in the oven.	• Pre-heat oven. • Steam made in the oven stops food drying, tinfoil & greaseproof paper also used. • Avoid opening oven door.	• Bread. Cakes. Biscuits. Pastries. • Vegetables e.g. potatoes, tomatoes, peppers. • Fruit e.g. Apples. • Puddings, apple crumble.	• Attractive appearance. • Doesn't add fat. • Little loss of nutrients. • Dries food.



GRILLING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Fast method of cooking food by radiant heat under a grill. 	•Pre-heat grill. •Seal surface of the food with high heat, preventing nutrient loss. •Use tongs to turn food. •Thin pieces of food only. •Don't salt food before grilling. •Oil grill grid to prevent sticking.	•Meat - chops, steaks, burgers, rashers. •Vegetables - tomatoes. •Fish fillets.	• Fast. •Little loss of nutrients. •Reduces fat content. •Attractive appearance & taste. •Constant attention needed. •Dries food. •Spatters grease. •Unsuitable for tough or thick cuts of meat.

BARBECUING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking food by radiant heat on a grid over glowing charcoal. 	•Pre-heat barbecue. •Seal surface with high heat, preventing nutrient loss. •Use tongs to turn food. •Thin pieces of food only. •Don't salt food before barbecuing. •Oil barbecue grid to prevent sticking.	•Meat, chops, steaks, burgers, rashers. •Vegetables e.g.tomatoes •Fish fillets.	• Fast •Little loss of nutrients. •Reduces fat content. •Attractive appearance & taste. •Constant attention. •Dries food. •Spatters grease.

ROASTING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
•Cooking food in a little fat in (a) a roasting tin in the oven, (b) a saucepan on the hob (pot roasting), (c) on a spit, under a grill or in an oven (spit roasting).	• Pre-heat oven. • Cooking time depends on weight. •Baste every 30 minutes. •Use cooking juices for gravy. •Quick roasting (tender cuts) - 230°C for 20 mins, 190 °C for remainder. •Slow Roasting (less tender cuts) - 175 °C for all the cooking time.	•Meat - beef, lamb, pork. •Poultry. •Vegetables e.g. potatoes	•Vegetables & meat cooked together. •Cooking liquid used as gravy. •Attractive appearance & taste. •Not much prep. •Spatters grease. •Not suitable for tough cuts. •Greasy. •Shrinkage occurs. •Adds fat.



FRYING

Definition	Guidelines	Suitable foods	Advantages & Disadvantages
• Dry - cooking food by conduction in frying pan on the hob. • Shallow - cooking food in hot fat. • Deep - cooking food immersed in hot fat.	• Pre-heat oil. • Use tongs for turning. • Drain before serving. • Coating food: • Protects surface. • Texture. • Flavour. • Appearance. 	• Meat - thin pieces e.g. rashers, chops, burgers, sausages. • Fish - thin pieces e.g. steaks, cutlets, fillets. • Eggs. • Vegetables. • Doughnuts.	• Quick. • Attractive taste & appearance. • Constant attention. • Spatters grease. • Not suitable for tough meats or large cuts. • Not suitable when cooking for large numbers. • Greasy. • Indigestible

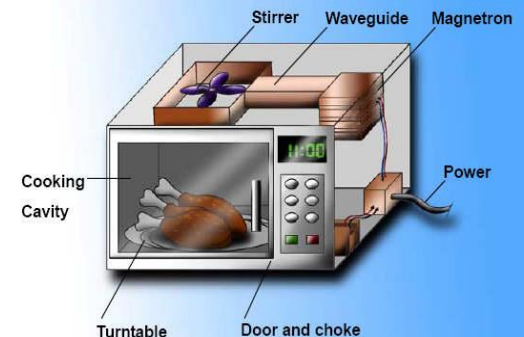
PRESSURE COOKING

Definition & principle	Structure	Guidelines	Suitable foods
•Fast moist method of cooking food @ high temps. in a pressure cooker. •Used for boiling, stewing & steaming. •Water under increased pressure boils @ higher temp. Pressure increased inside pressure cooker by not letting steam escape. As a result, water boils @ 122°C. Food cooks in 1/3 of normal time.	•Heavy gauge saucepan. Locking lid. •Rubber gasket seals lid. •Trivet. •Separator basket. •Vent on lid. •Safety valve.	•Follow instructions. •Never overfill. •Build up steam before applying weight. •Time very accurately. •Reduce pressure before opening- •(a) Stand for 15 mins @ room temp. •(b) Under cold running water.	•Stocks & soups. •Stews. •Vegetables e.g. potatoes. •Jam. •Christmas pudding.



MICROWAVE

Principle	Guidelines	Suitable dishes
•Electro-magnetic waves penetrate the food to a certain depth and cause the molecules to vibrate very rapidly. This causes intense heat. The remainder of the food cooks by conduction.	•Time carefully. •Timing is affected by: Composition. Thickness. Density. Temperature at the start. Amount of food. •Cover food, keep moist, cook faster. •Turn or stir. •Pierce skins. •Arrange in circle, thickest bit outward. •No metal or foil. •Allow standing time.	•Defrosting. •Reheating. •Cooking. •Melting chocolate. •Heating liquid.
		•Unsuitable dishes •Pastry. •Meringue.



IMPORTANCE OF COOKING FOOD

1-Cooking makes food easy to digest.

2-Cooking improves the appearance, texture, colour, flavour, and taste of food .

3-Cooking of food adds variety to the our meals.

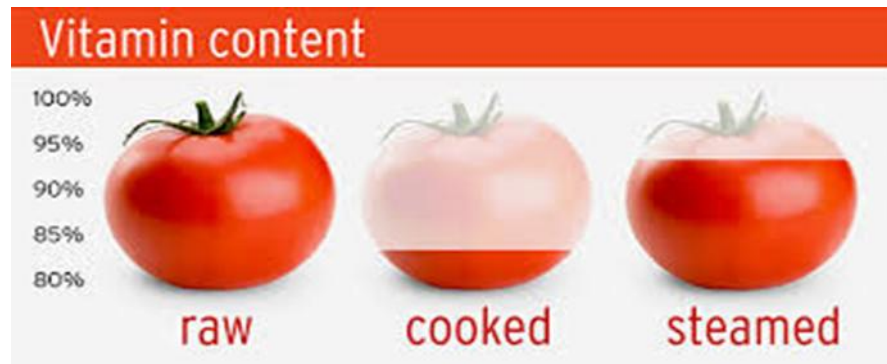
4- Cooking helps to keep food longer and longer shelf life.

5- Cooking makes food safe from micro-organisms are present in raw foods.

LOSSES OF NUTRIENTS DURING COOKING

1- Vitamin A:

- It gets oxidized when it reacts with oxygen present in the air and gets destroyed.
- Cooking at high temperature in an open pan results in food coming in contact with oxygen for a long period of time and this brings about a reduction in vitamin A content.



LOSSES OF NUTRIENTS DURING COOKING

2-Vitamin B Complex

- It is a group of eight water soluble vitamins.
- Vitamin B gets dissolved in water when these foods are washed, soaked or cooked in water.
- If this water is discarded, it results in the loss of Vitamin B.

3- Vitamin C

- It is another water soluble vitamin which is easily destroyed by heat and oxidation.
- Vitamin C is also lost when you wash vegetables and fruits after cutting or if you cut them too fine.

LOSS OF NUTRIENTS DURING COOKING



LOSSES OF NUTRIENTS DURING COOKING

4-Proteins

Cooking results in softening of proteins in foods such as egg, fish and meat.

- All proteins present in the food items absorb water and get coagulated by heat, it loses moisture and becomes dry and rubbery.
- Addition of acidic ingredients like lemon juice, tomatoes, curd increases cooking time and makes the protein tough and leathery.

LOSSES OF NUTRIENTS DURING COOKING

5-Minerals

- Minerals like sodium and potassium dissolve in water.
- Minerals get lost when food is first cut and then washed and the extra water in which they are boiled, is thrown away.



How to Prevent nutrient loss during cooking

- 1-Keep skins on when possible
- 2-Avoid continuous reheating of food
- 3-Use a minimal amount of cooking liquid
- 4-Choose steaming over boiling
- 5-When you do boil, retain the cooking liquid for a future use (like soups and stocks)
- 6-Use a pressure cooker when possible
- 7-Cut veggies into large chunks to reduce surface area

THE FACTORS INFLUENCING FOOD INTAKE

7- Food Characteristics

6- Serving temperature :

-Food temperatures preferred by individuals vary with age and other personal factors . Both **hot** and **cold** temperatures are preferred for different types of foods .

lecture 4

**FACTOR TO BE CONSIDERED IN
MENU PLANNING**

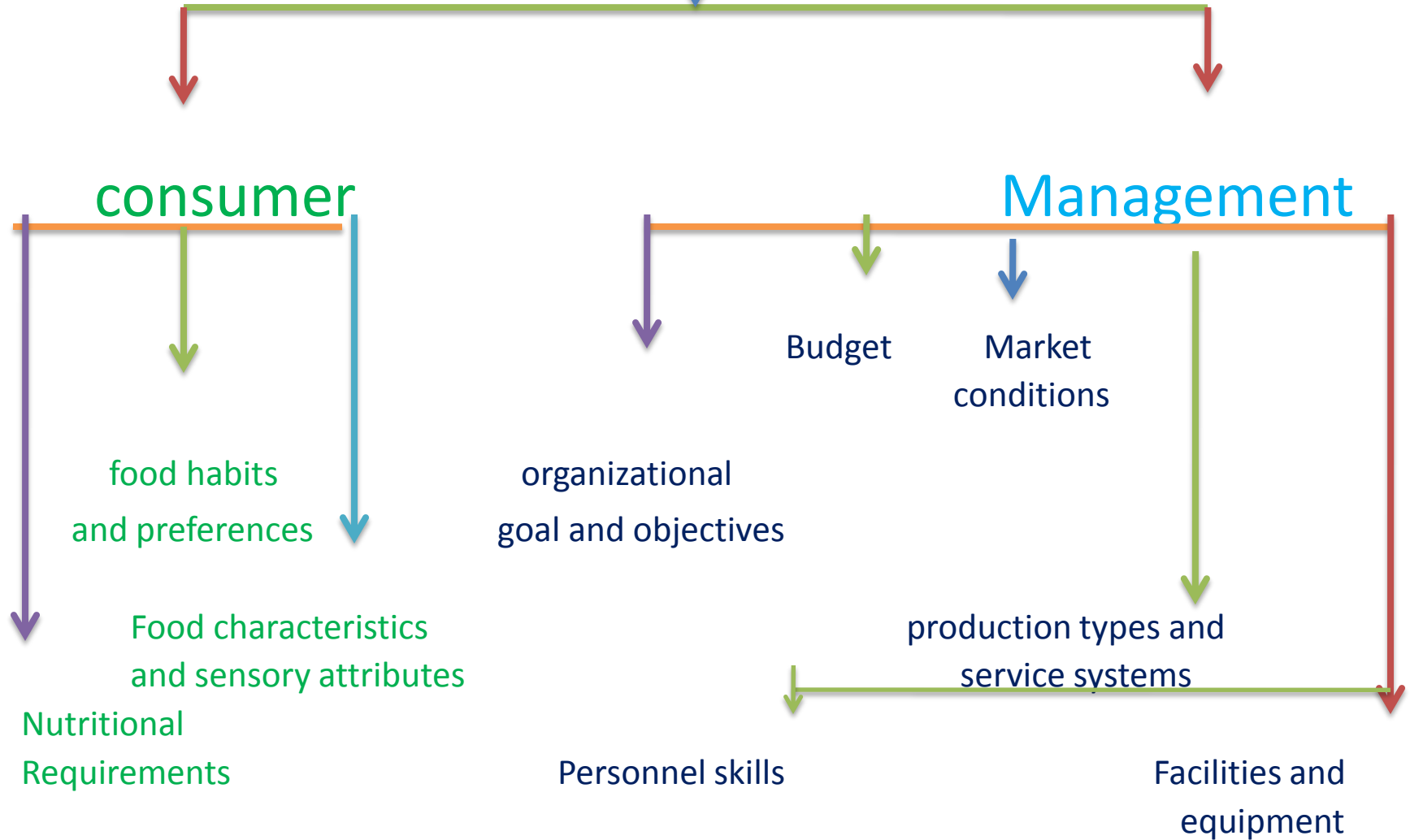
FACTOR TO BE COSIDERED IN MENU PLANNING

1- Management's Factors

2- Consumer's Factors

FACTOR TO BE COSIDERED IN MENU PLANNING

Menu



FACTOR TO BE CONSIDERED IN MENU PLANNING

1- management's factors

1- Organizational goals and objectives:

This factor is very important in menu planning . For example the menu planned for health club and hospital will be different from that of a restaurant.

2- Budget :

This factor depends on the amount of money that can be spent depends on several factors , such as the (income from food sales and the relative food cost percentage .)

FACTOR TO BE COSIDERED IN MENU PLANNING

1- management's factors

3- Market conditions : This factor depends on seasonal availability of foods products

4- Physical facilities and equipment :

5- Personnel skills :

6- Types of production and service systems:

FACTOR TO BE CONSIDERED IN MENU PLANNING

2- Consumer's Factors

1-Food habits and preferences

2- Food characteristics and sensory attributes

3- Nutritional requirements



2- Consumer's Factors

1-Food habits and preferences

2- Food characteristics :

A- Color

B- Texture and shape

C- Consistency

D- Flavor

E- Method of food preparations

F- Serving temperature

G- Presentation .

H- Method of cooking.



2- Consumer's Factors

3- Nutritional requirements :

This is the most important factor in menu planning . At some food service operations , it may be mandated regulatory agencies. Meeting nutritional needs in such institutions as hospitals, nursing home, and schools must be given top priority when planning menu .

lecture 4 (CONT)

Nutritional requirements

Nutritional requirements

For example :

Special nutritional requirement

- 1- Pregnancy**
- 2- Lactation**
- 3- Infancy**
- 4- Childhood**
- 5- Adolescence**
- 6- Old age**



Special Nutritional requirements

1- Pregnancy :

energy	Proteins	Minerals	Vitamins
The energy needs increase during the second and third Stages of pregnancy. Additional 300 kcal take care of the increased demands of pregnancy . 	The proteins requirement increases by 15 g for the synthesis of foetal and maternal tissues. 2 g protein / 1 kg idle body weight .	The calcium requirement is (1 g and iron is 38 mg for the formation of foetal bones and teeth and formation of blood, iodine in the form of iodized salt, help in protecting the mother and child against goitre, cretinism.	With an increase in calories the need for vit. B1, B2, and niacin increase as they are needed for release of energy from carbohydrates , proteins , fats, the need for folic acid and vit. D also increase .

Special Nutritional requirements

2- Lactation :

Energy	Proteins	Minerals	Vitamins
Calorie requirement increase by 550 kcal during the first six months , followed by a marginal decrease during the next six months with 400 kcal being to meet the additional demands .	An additional intake of 25 g in the first six months and 18 g in the next six months is adequate .	Calcium requirement is 1,000 mg which is necessary for synthesis of milk. Additional iron is not prescribed as milk is poor source of iron .	The requirement for vitamins A, D, B- complex and vit. C increases . Vit. A =950 ug. Vit. C= 80 mg . Vit. B1,2 = 1,5 ug. Folic acid = 150 ug.



Special Nutritional requirements



3- Infancy :

During the first few day's after delivery **colostrums is secreted which is not mature milk but the substance richer in proteins and vitamin A, and antibody's .**

Human milk is the best food for infants but it deficient in **iron , vitamin C, and D . And we should be given supplementary foods .**

3- Infancy

Supplementary foods during first year :

The age at which supplementary foods should be included depends on a number of factors such as literacy of mother , economic factor, and time available. Supplements if not prepared, stored, and fed to the infant in hygienic conditions is one of the main causes for diarrhoea and gastro- intestinal upsets.

3- Infancy

points should be borne in mind While introducing a new food the following :

- 1- Give a teaspoonful in the beginning .**
- 2- Never force feed an infant .**
- 3- Introduce one food at a time till the infant's system is used to the new food . Season with salt**

3- Infancy

points should be borne in mind While introducing a new food the following :

- 4- Start with a thin smooth consistency initially**
- 5- Once the infant is used to supplements, introduce variety .**
- 6- Follow hygienic practices and do not give any leftover food or drink to an infant .**



3- Infancy

Choice of food supplements for infants

- 1- Egg yolk**
- 2- Strained cereals**
- 3- Strained soup**
- 4- Rice**
- 5- Pureed vegetables**
- 6- Ready to eat cereals**
- 7- Fruits and fruit juices**



Special Nutritional requirements

4- Childhood

This stages rapid growth takes place .

Nutritional requirement for **Childhood :**

1- Energy :

More energy per kg body weight is required for .

A- Rapid growth .

B- Enhanced physical activity .

C- High BMR as compared to an adult .



4- Childhood

2- Proteins :

- ❖ **Good quality proteins should be included to take care of body building and maintenance of tissues.**
- ❖ **Milk proteins are complete protein and provide calcium as well .**

3- Carbohydrates and fats :

- ❖ **They provide calories and spare proteins from being oxidized for energy .**
- ❖ **Refined carbohydrates and poor dental hygiene are the reasons for dental caries in children.**
- ❖ **The consumption of fruits and vegetables, both cooked and uncooked , should be encouraged.**
- ❖ **Junk food and beverages should be discouraged.**

4- Childhood

4- Vitamins and minerals :

If the diet is well planned keeping the principles of menu planning in mind supplements may not be necessary .



4- Childhood

Points to be considered during feeding Childhood

- 1- Regular meal timings.**
- 2- No nibbling low nutrient dense snacks between meals**
- 3- Do not force feed let the normal appetite return.**
- 4- Meal timings should be pleasant .**



Points to be considered during feeding Childhood (cont.)

- 5- Food should be appetizing and attractively served. Finger foods should be preferred for children.**
- 6- Packed meals should not be messy to eat.**
- 7- Attractive colors and shapes appeal to children and with a little imagination eating green vegetables and drinking milk could become more interesting .**

Special Nutritional requirements

5- Adolescence :



Adolescence is a period of physiological stress for the body because of extremely rapid rate of growth.

Changes during this Adolescence stages

- 1) The BMR is increases and other hormonal changes which take place .**
- 2) Girls grow rapidly than boys .**
- 3) A deficiency of calcium and vit. D during infancy or childhood results in rickets.**
- 4) Adolescence depended on snacks and junk food.**

5- Adolescence

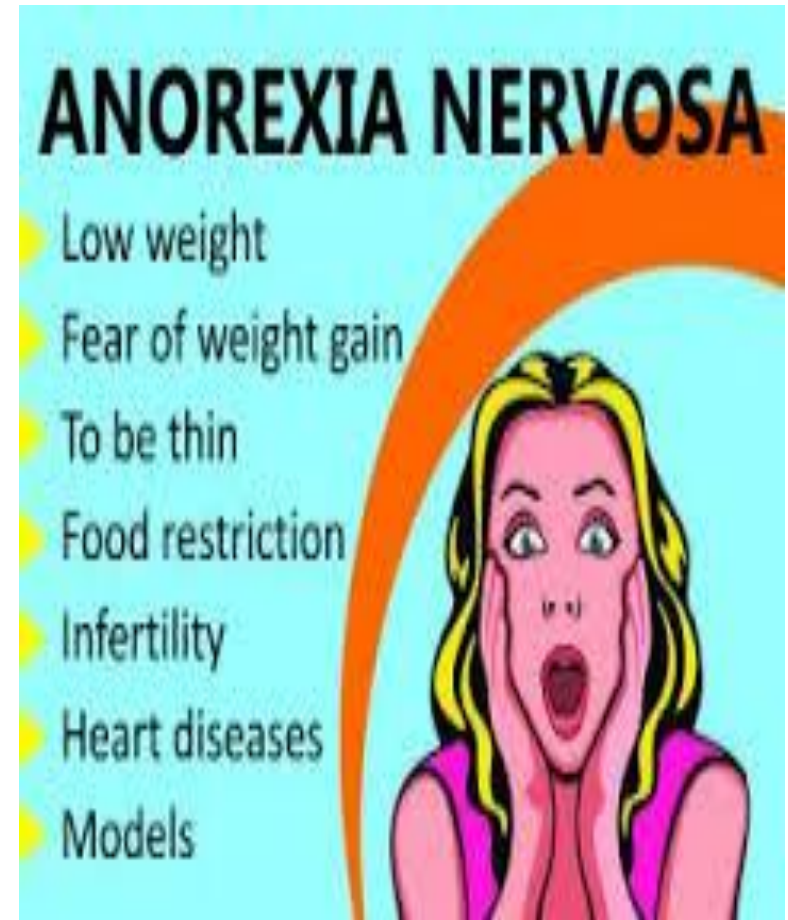
**In general the diet is deficient in the protein food group and protective food group. Food such as :
Milk , green leafy vegetables , yellow fruits and vegetables citrus fruits , and whole grain cereals are deficient in the diet of adolescents .
Deficiency during infancy and childhood may leave their problems in adolescents .**



Problems during Adolescence

A – Anorexia nervosa :

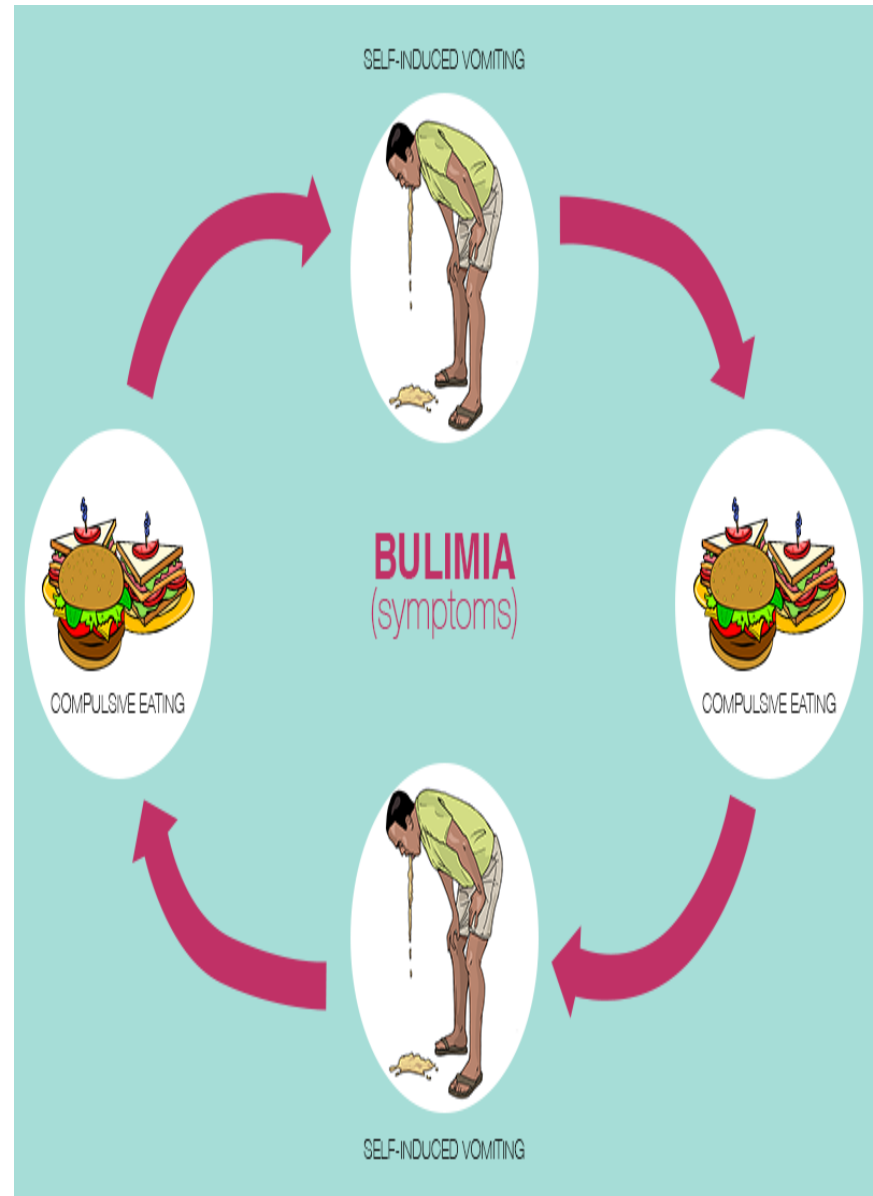
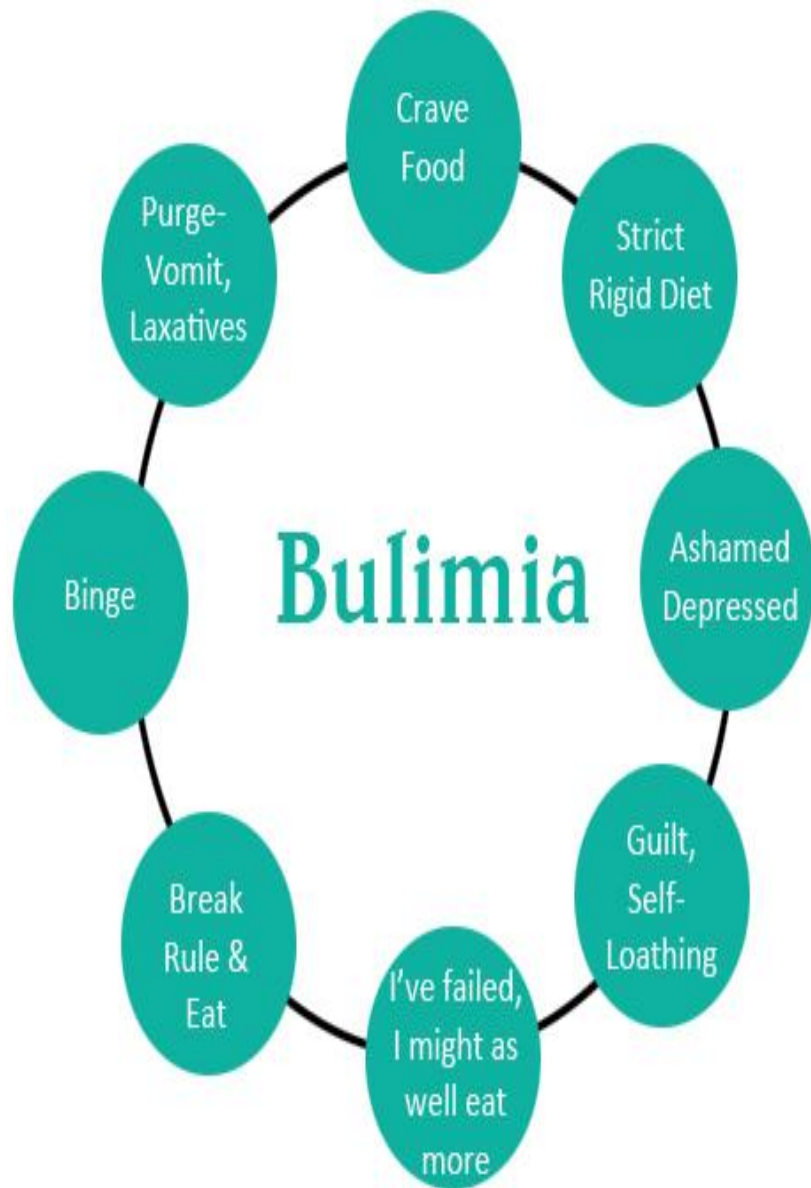
This is a disorder seen in adolescent girls lead to , weight loss, loss of muscle tissues , a low BMR and serious health problems.



5- Adolescence

B- Bulimia :

It is another disorder in which the appetite increases drastically followed by induced vomiting to throw out whatever has been consumed . Both disorder are mainly seen in adolescent girls from affluent families.



Modifications in diets for Children and Adolescents

Preschool children	Children going school	Adolescent
• A high calorie high protein diet, rich in calcium and vitamin A. • - Mildly flavored and less spicy foods to be given • - Handy Finger Foods are preferred. Example French fries, Sandwiches, vegetable rolls, etc. • - Foods should neither be too hot or too cold for the child to handle	• A high calorie, high protein diet with plenty of vitamins and minerals. • Need energy rich foods for their activities both at school and home. • Packed meals assumes a lot of importance as break fast is usually skipped.	• A high calorie, high protein diet, rich in calcium and iron. • Quantity of food in take must be in- creased to meet their rapidly changing body needs. • Nutritious fast foods and snacks should be planned. • Starving crash diet- must be discouraged.

Special Nutritional requirements

6- Old age

Types of changes are seen in old age

A- Body composition

C- Sensory changes

B- Digestive functions

D- Metabolic changes

Types of changes in old age

A- Changes in body composition for old age :

1- Lose muscle mass

2- gain fat.

3- Active muscle is lost.

4- BMR reduces .

5- Bone loss occurs lead to reduction in weight.

6- Collagen content increases giving rise to loss of elasticity in blood vessels, joints, and skin .



Types of changes in old age

B- Changes in digestive functions for old age

- 1- Loss of teeth and ill- fitting dentures make it necessary to serve soft well cooked food .**
- 2- Decreased secretion of saliva makes food difficult to swallow.**
- 3- Heartburn , diarrhoea, or constipation are other complaints.**

Types of changes in old age

C- Sensory changes for old age :

The five senses of sight, hearing, taste, smell, and touch become dim with age.

D- Metabolic changes in old age :

- 1-Changes in metabolism specially (glucose tolerance, and anaemias are sometimes seen).**
- 2- Osteopenia and osteoporosis are seen .**

Nutritional requirements for Old age :

The main features (characteristic) of the diet for old age

- 1- A low calorie diet of 1,000 – 1,500 kcal .**
- 2- Good quality protein, calcium, and iron. Vit. A, C, and folic acid to prevent anaemia .**
- 3- Hot meals should be served attractively as appetite is low .**
- 4- Food should be easy to chew,**

Nutritional requirements for Old age :

The main features (characteristic) of the diet for old age

- 5-liquid preparations should be included in each meal as they are light and easy to digest.**
- 6- Five small meals should be preferred to a three meals pattern .**
- 7- Gas forming foods and strongly flavored vegetables such as cauliflower, cabbage, and onion should be avoided.**

Nutritional requirements for Old age :

The main features (characteristic) of the diet for old age

- 8- Fiber rich foods (such as whole grains, green leafy vegetables, pulses, and fruits) are nutritious and provide fiber necessary for normal elimination . They should be added into the diet gradually to prevent gastric irritation and gas.**
- 9- Fluids : Increasing fluids are necessary to flush out toxic wastes. The kidneys function at a slower pace with age, but fluids help in excretion of toxins.**
- 10- Low calories density and high satiety value foods should be included.**